

ELEC5305 Project Proposal

Automatic Speech–Music Classification Using Audio Features and STFT

The University of Sydney
ELEC5305 – Acoustics, Speech and Signal Processing

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GitHub Repository:
<https://github.com/49sky/elec5305-project-530582937>

1. Project Overview

This project will develop a system that classifies a short audio recording as either speech or music. Automatic audio classification is useful for organising large media collections, broadcast monitoring and audio retrieval. The project will use signal-processing methods from ELEC5305 rather than a large end-to-end neural network. Audio will be divided into short frames, analysed with the Short-Time Fourier Transform (STFT), and described using simple time-domain and spectral features. These features will then be used by a supervised classifier. The main goal is to build a working MATLAB prototype and identify which features and STFT settings give reliable speech–music separation.

2. Background and Motivation

Audio classification is useful because large amounts of speech and music are recorded, streamed, and stored every day. Being able to automatically identify the type of audio can support applications such as media organization, search, content monitoring, and speech-based systems.

This topic was chosen because it is a practical way to apply the main ideas introduced in ELEC5305. The course focuses on understanding audio signals, analyzing their characteristics, and using signal-processing methods in MATLAB. The Week 3 material also introduces audio content analysis and shows that different sounds can be described using measurable features such as energy, zero-crossing rate, and spectral characteristics.

Speech and music often have different patterns over time and frequency, so these features can be used to help distinguish between them. This project will therefore investigate whether a small group of simple audio features can provide reliable speech–music classification without using an overly complex system. Previous studies show that combining several low-level audio features can separate speech and music effectively (Scheirer & Slaney, 1997; Tzanetakis & Cook, 2002).

3. Proposed Methodology

The project will be developed in MATLAB using a collection of speech and music recordings.

First, the audio files will be prepared so that they have the same format and sampling rate. Each recording will then be divided into short sections so that its characteristics can be analyzed over time.

Next, several simple audio features introduced in ELEC5305 will be calculated. These may include RMS energy, zero-crossing rate, spectral centroid, spectral flux and spectral rolloff. The Short-Time Fourier Transform (STFT) will also be used to view how the frequency content of the audio changes over time.

The extracted features will then be used as inputs to a simple classification method, such as a decision tree or support vector machine, which will learn the differences between speech and music from a set of labelled examples.

Finally, the completed system will be tested using audio recordings that were not used during training. The results will be evaluated using classification accuracy and a confusion matrix. Different combinations of audio features may also be compared to determine which features are most useful for separating speech from music.

4. Expected Outcomes

The expected result is a MATLAB program that accepts an audio file and predicts whether it is speech or music. Performance will be reported using classification accuracy, precision, recall, F1-score and a confusion matrix. The final report will also discuss which features are most useful, how window length affects the result and where the classifier makes mistakes. The project will therefore demonstrate both practical audio classification and understanding of the signal-processing methods used to build it.

5. Timeline

Week	Task
Weeks 1–2	Confirm project topic, scope and GitHub setup.
Weeks 3–5	Literature review, collect dataset and prepare data .
Week 6-8	Implementations
Week 9-11	Training and testing
Week 12	Prepare results, figures and draft report.
Week 13	Final report, GitHub documentation and demonstration.

6. References

- Scheirer, E. D., & Slaney, M. (1997). Construction and evaluation of a robust multifeature speech/music discriminator. *Proceedings of ICASSP*, 2, 1331–1334. <https://doi.org/10.1109/ICASSP.1997.596192>
- Tzanetakis, G., & Cook, P. (2002). Musical genre classification of audio signals. *IEEE Transactions on Speech and Audio Processing*, 10(5), 293–302. <https://doi.org/10.1109/TSA.2002.800560>
- Giannakopoulos, T. (2015). pyAudioAnalysis: An open-source Python library for audio signal analysis. *PLOS ONE*, 10(12), e0144610. <https://doi.org/10.1371/journal.pone.0144610>

Snyder, D., Chen, G., & Povey, D. (2015). MUSAN: A music, speech, and noise corpus. arXiv:1510.08484.